Question ID 0bd33265

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0bd33265

The equation $h = \frac{9(v-273.15)}{5} + 32$ gives the corresponding temperature h, in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of 467.33 degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

ID: 0bd33265 Answer

Correct Answer:
515

Rationale

The correct answer is 515. It's given that the equation $h = \frac{9(v-273.15)}{5} + 32$ gives the corresponding temperature h, in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. Substituting 467.33 for h in the given equation yields $467.33 = \frac{9(v-273.15)}{5} + 32$. Subtracting 32 from both sides of this equation yields $435.33 = \frac{9(v-273.15)}{5}$. Multiplying both sides of this equation by 5 yields $2{,}176.65 = 9(v - 273.15)$. Dividing both sides of this equation by 9 yields $241.85 = v - 273.15$. Adding 273.15 to both sides of this equation yields $515 = v$. Therefore, if a substance has a temperature of 467.33 degrees Fahrenheit, the corresponding temperature, in kelvins, of this substance is 515.

Question Difficulty:
Hard